Lecture 14: Standard Errors and Weak IV

Jeremy Majerovitz (University of Notre Dame)

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Simulation

For each simulation:

$$Y_i = \alpha + \beta \cdot X_i + \varepsilon_i$$
$$X_i = \gamma + \pi \cdot Z_i + u_i$$

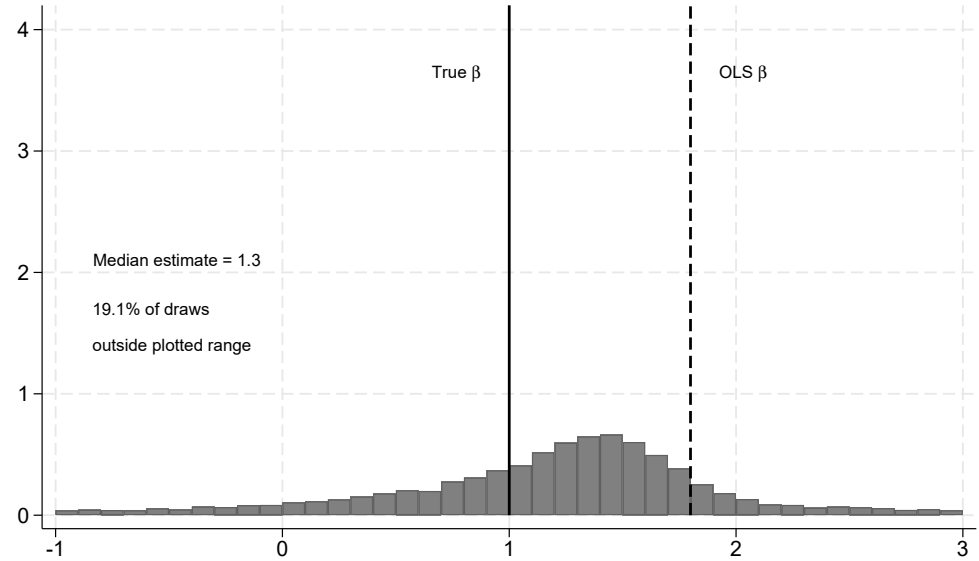
- Z_i,u_i, and ε_i all $\sim N(0,1)$, and jointly normal
- $\text{Corr}(u_i, \varepsilon_i) = 0.8$; Z_i is uncorrelated with u_i and ε_i
- Each observation is drawn i.i.d., $N = 1,000$
- Set $\beta = 1$ and $\alpha = \gamma = 0$ (intercepts don't matter for simulation)
- Note that X_i is endogenous: OLS coefficient is biased upward

Across simulations, we vary π to achieve different first-stage strength, $\mathbb{E}[F]$

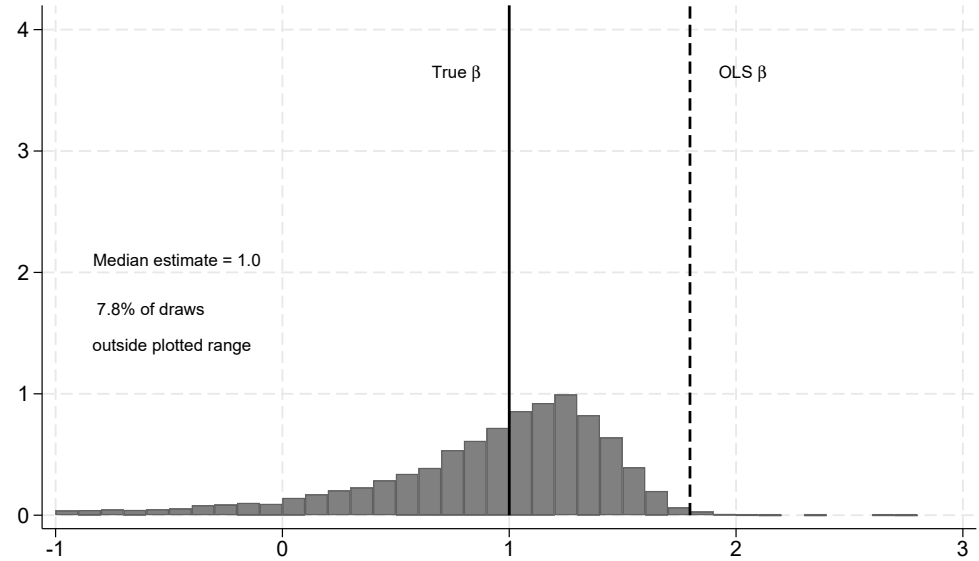
Repeat simulation 10,000 times

Plot distribution of $\hat{\beta}^{IV}$ across simulations

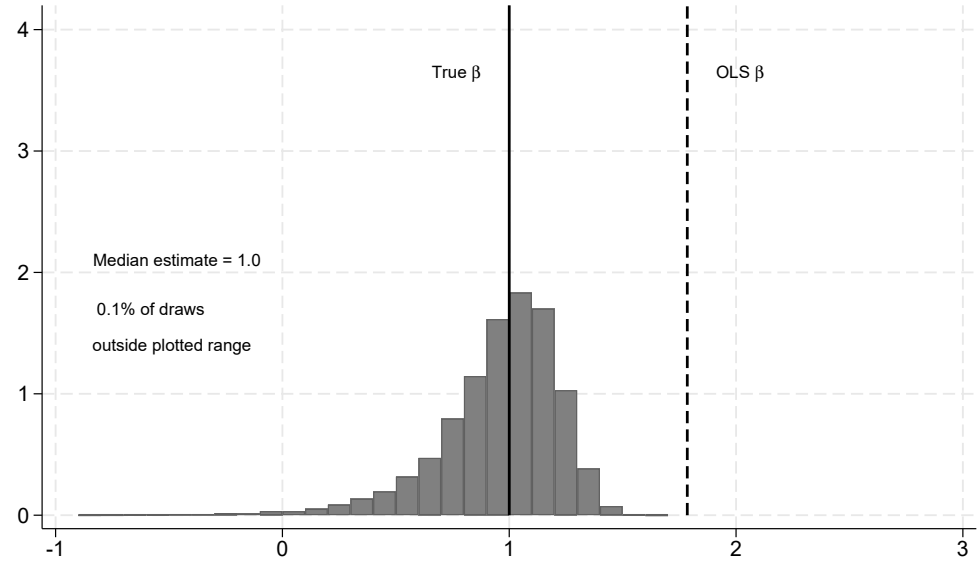
Distribution of $\hat{\beta}^{IV}$, $\mathbb{E}[F] = 2$



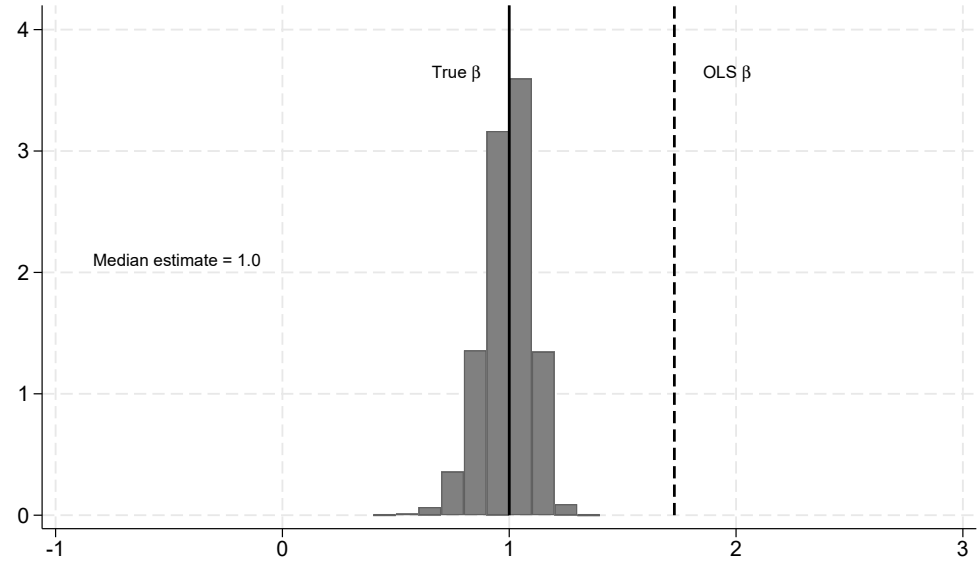
Distribution of $\hat{\beta}^{IV}$, $\mathbb{E}[F] = 5$



Distribution of $\hat{\beta}^{IV}$, $\mathbb{E}[F] = 20$



Distribution of $\hat{\beta}^{IV}$, $\mathbb{E}[F] = 100$



Hypothesis Testing

Next, we test the null hypothesis $\beta = 1$

Construct t-statistic:

$$t = \frac{\hat{\beta}^{IV} - 1}{\widehat{\mathsf{SE}}\left(\hat{\beta}^{IV}\right)}$$

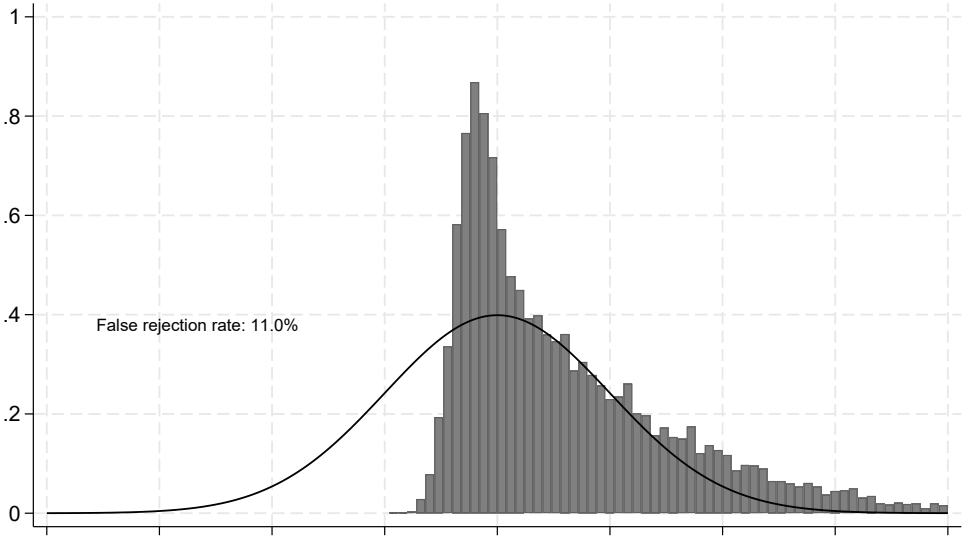
Repeat simulation 10,000 times

Compare to pdf of the standard normal distribution

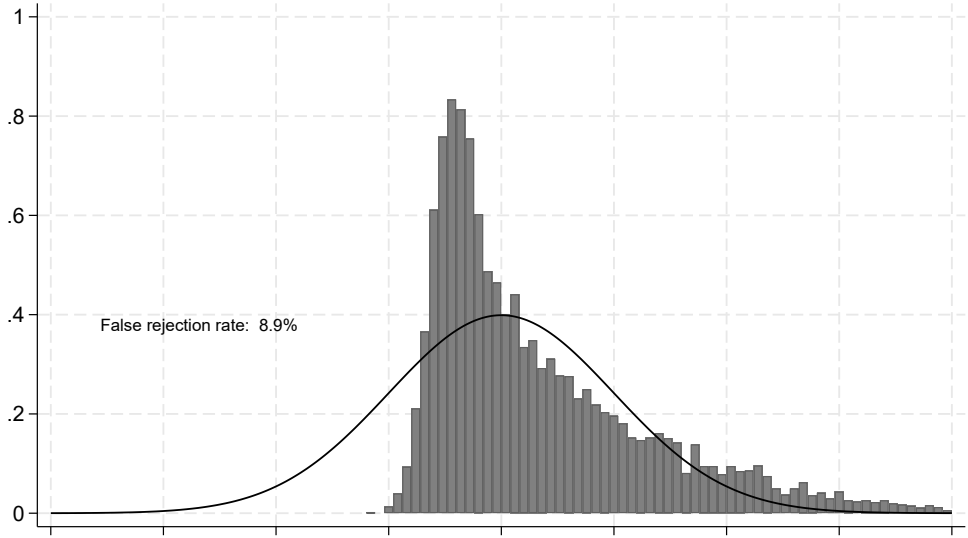
Measure false rejection rate with standard-normal critical values

- Should be 5% for a 5% test!

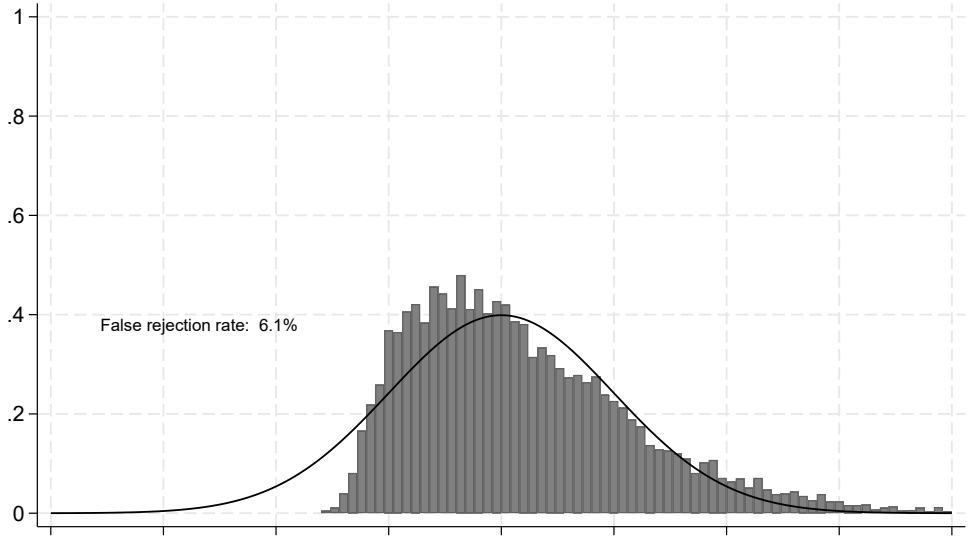
Distribution of $\frac{\hat{\beta}^{IV}-1}{\widehat{\mathbf{SE}}(\hat{\beta}^{IV})}$, $\mathbb{E}[F]=2$



Distribution of $\frac{\hat{\beta}^{IV}-1}{\widehat{\mathbf{SE}}(\hat{\beta}^{IV})}$, $\mathbb{E}[F]=5$



Distribution of $\frac{\hat{\beta}^{IV}-1}{\widehat{\mathbf{SE}}(\hat{\beta}^{IV})}$, $\mathbb{E}[F] = 20$



Distribution of $\frac{\hat{\beta}^{IV}-1}{\widehat{\mathbf{SE}}(\hat{\beta}^{IV})}$, $\mathbb{E}[F] = 100$

